

DAYANANDA SAGAR COLLEGE OF ENGINEERING

**Shavige Malleshwara Hills, Kumaraswamy Layout, Bangalore-560078
Department of Artificial Intelligence and Machine Learning**



INTERNSHIP REPORT

ON

“Associate System Architect Intern”

Submitted in partial fulfillment for the award of degree(20AI8ICINT)

BACHELOR OF ENGINEERING IN

Department of Artificial Intelligence and Machine Learning

Submitted by:

ABHIJIT PATTANAİK - 1DS20AI002

Conducted at



Pega Systems

**Department of Artificial Intelligence and Machine Learning
Dayananda Sagar College of Engineering Bangalore-78**

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Department of Artificial Intelligence and Machine Learning



CERTIFICATE

This is to certify that the Internship titled “**Associate System Architect**” carried out by **Mr. Abhijit Pattanaik** bearing USN **1DS20AI002**, a bonafide student of Dayananda Sagar College of Engineering, in partial fulfillment for the award of **Bachelor of Engineering**, in **Artificial Intelligence and Machine Learning** during the year 2023-2024. It is certified that all corrections/suggestions indicated have been incorporated in the report.

The internship report has been approved as it satisfies the academic requirements in respect course Internship (**20AI8ICINT**)

**Signature of Reporting
Head**

Satish Babu
Soma Sridhar
Sr. Manager
PegaSystems

**Signature of Internship
Coordinator**

Prof. Kavya D N
Dept. of AI & ML
DSCE

Signature of HoD

Dr. Vindhya P
Malagi Dept. of AI &
ML DSCE

External Viva:

Name of the Examiner

1) _____

2) _____

Signature with Date

DAYANANDA SAGAR COLLEGE OF ENGINEERING

Shavige Malleshwara Hills, Kumaraswamy Layout, Bangalore-560078

Department of Artificial Intelligence and Machine Learning



DECLARATION

I, **Abhijit Pattanaik**, final year student of Artificial Intelligence and Machine Learning, Dayananda Sagar College Of Engineering, Bangalore - 560078, declare that the Internship has been successfully completed, in **Pega Systems**. This report is submitted in partial fulfillment of the requirements for the award of Bachelor Degree in Artificial Intelligence and Machine Learning, during the academic year 2023- 2024.

Date:

Place: Bangalore

NAME: ABHIJIT PATTANAIAK

USN: 1DS20AI002

OFFER LETTER



PEGASYSTEMS WORLDWIDE INDIA PRIVATE LIMITED

Registered Office: Building No.12A, 13th Office Level,
MindSpace Cyberabad, Madhapur, Hyderabad -500081

CIN No. U72200TG2007FTC055018

Abhijit Pattanaik
B3, Jubilee Square, 70th Cross Road, Near Vasavi Hospital, Kumaraswamy Layout, Bengaluru, Karnataka, 560078.

Dear Abhijit,

Congratulations!

Pegasystems is pleased to offer you the role of Intern as part of our 2024 Internship Program. You will be paid a monthly stipend of INR 25,000, subject to tax and statutory deductions, for a maximum 40-hour work week. Pega policy does not require interns to work greater than 40 (forty) hours per week. The duration of the internship is for a period of 4 (four) months.

The purpose of an internship is to provide students with the opportunity to gain valuable business skills while earning a competitive compensation. Intern positions do not include any health coverage or other corporate benefits. As part of the internship, you will be asked to abide by all policies, including the company Code of Conduct and completing a weekly time sheet.

Please sign and return your signed Offer Letter and agreement no later than 20th February 2024.

We look forward to working with you and hope you enjoy your time at Pegasystems!

Sincerely,
For Pegasystems Worldwide India Private Limited

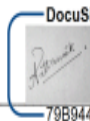
DHARANIKOT Digitally signed by
A SUYODHAN DHARANIKOTA SUYODHAN
Date: 2024.01.22 17:20:01
+05'30'

Authorized Signatory

I accept the terms of this offer letter and will begin an Internship at Pegasystems.
I confirm the start date is 20th February'2024

I understand and acknowledge that this document has been authenticated in electronic mode and such signing or delivery of the document shall be deemed to be as good as an original document and not affect the validity of provisions it contains.

DocuSigned by:



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22 January 2024 | 08:06 EST

Abhijit Pattanaik

Date

COMPLETION CERTIFICATE

ACKNOWLEDGEMENT

This Internship is a result of accumulated guidance, direction and support of several important persons. We take this opportunity to express our gratitude to all who have helped us to complete the Internship.

We express our sincere thanks to our Principal Dr. B G Prasad, for providing us adequate facilities to undertake this Internship.

We would like to thank Dr. Vindhya P Malagi, Head of Department – AI & ML, for providing us an opportunity to carry out Internship and for her valuable guidance and support.

We would like to thank Mr Dinesh Bherwani, PegaSystems ,for guiding us during the period of internship.

We express our deep and profound gratitude to our Internship Co-ordinator, Prof. Kavya D N, Assistant Professor – AI & ML, for her keen interest and encouragement at every step in completing the Internship.

We would like to thank all the faculty members of AI & ML department for the support extended during the course of Internship.

We would like to thank the non-teaching members of the Department of AI & ML, for helping us during the Internship.

Last but not the least, we would like to thank our parents and friends without whose constant help, the completion of Internship would have not been possible.

ABHIJIT PATTANAIK
1DS20AI002

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CHAPTER 1

INTRODUCTION

I am currently interning at PegaSystems as an Associate System Architect Intern from February 20, 2024. Pegasystems is a software company that specializes in customer relationship management (CRM) and business process management (BPM) solutions. It was founded in 1983 and is headquartered in Cambridge, Massachusetts, USA. Pega's software is used by businesses in various industries to automate and streamline their operations, improve customer engagement, and drive better outcomes. The company's flagship product is the Pega Platform, which provides a unified platform for building and deploying enterprise applications. Pega's customers include some of the world's largest and most successful companies in industries such as financial services, healthcare, telecommunications, and government. The company has a global presence with offices in more than 30 countries and serves customers in over 140 countries. During the tenure of my internship, I am involved in learning the Pega Platform and its various use cases to develop an application with the help of Pega Community Edition. The application installs a 20GB software Pega Platform: Community Edition, that is used by Pega Developers to build CRM applications directly by communicating with the clients. Along with the learning, I was also involved in the System Architect Training course offered by the company which covered the basics of Pega Platform to work on it. Some of the concepts are Rules, Rulesets, Activity, Data Models, Data Pages, etc. Based on the training I had to clear the Pega System Architect Certification Exam with a score of at least 65% which I cleared with 74%. After the CSA (Certified System Architect) Training and Certification, I was trained to develop my skill set for the Senior System Architect which covered the Pega Concepts on a deeper level and its certification exam is still remaining to be cleared. Once all the training was finished, I also worked on a Group Based Sample Application to build an Effective Key Management Application.

CHAPTER 2

DESCRIPTION OF ORGANIZATION

Pegasystems Inc., commonly referred to as Pega, stands out as a prominent American software company renowned for its focus on customer excellence solutions. Established in 1983 by Alan Trefler, its headquarters are nestled in the tech hub of Cambridge, Massachusetts. Pega's core offering revolves around its flagship product, the Pega Platform, which has evolved over the years to become a cornerstone in the realm of enterprise application development.

At the heart of Pega's offerings lies the Pega Platform, formerly known as PRPC (Pega RULES Process Commander). This platform serves as a robust foundation for organizations seeking to automate and streamline their business processes effectively. The distinguishing feature of the Pega Platform lies in its low-code development environment, designed to empower users to build and deploy sophisticated business applications with relative ease and efficiency, often with minimal coding expertise required.

One of the key advantages of the Pega Platform is its visual modeling approach, which fundamentally transforms the application development landscape. Through intuitive drag-and-drop functionality and a rich array of pre-built templates, Pega Platform users can swiftly design, customize, and implement complex business applications. This approach significantly accelerates the development lifecycle, allowing organizations to respond promptly to evolving business needs while minimizing the traditional reliance on intricate coding processes.

Moreover, Pega Platform's emphasis on agility and adaptability aligns seamlessly with modern business demands. By providing a versatile toolkit that caters to a diverse array of industries and use cases, Pega empowers organizations to rapidly innovate and iterate their applications in response to shifting market dynamics and customer preferences. As a result, Pega Platform emerges as a catalyst for digital transformation, enabling businesses to unlock new efficiencies, enhance customer experiences, and drive sustainable growth in an increasingly competitive landscape.

CHAPTER 3

SCOPE OF WORK

Custom Application Development:

Develop custom applications using Pega Platform to meet specific business requirements.

Utilize Pega's low-code capabilities for rapid application development and deployment.

Design and implement applications for various use cases such as CRM, BPM, case management, and more.

Legacy System Integration:

Integrate Pega Systems with existing legacy systems and third-party applications.

Ensure seamless data flow and interoperability between Pega applications and other systems.

Utilize Pega's integration capabilities to streamline business processes and improve efficiency.

Business Process Automation:

Automate manual and repetitive tasks using Pega Robotics and BPM capabilities.

Design and implement automated workflows to improve productivity and reduce operational costs.

Utilize Pega's AI-powered automation for intelligent decision-making and process optimization.

Customer Experience Enhancement:

Implement Pega Customer Decision Hub for personalized customer interactions.

Design and optimize customer journeys using Pega's customer engagement tools.

Enhance omnichannel customer experience through seamless integration with Pega applications.

Analytics and Reporting:

Utilize Pega's analytics capabilities for real-time insights and reporting.

Design and implement dashboards and reports to monitor key performance indicators.

Leverage predictive analytics for data-driven decision-making and business optimization.

Security and Compliance:

Implement robust security measures to protect sensitive data within Pega applications.

Ensure compliance with industry regulations and standards using Pega's compliance framework.

Conduct regular security audits and assessments to mitigate risks and vulnerabilities.

CHAPTER 4

LEARNING OBJECTIVES AS GIVEN BY COMPANY

During my internship at PegaSystems as an Associate System Architect, initially we were trained with the terms used in Pega Software and the working of it, and also how to create an application using Pega Platform with low coding. After the training, we were given a Sample App Project: Movie Ticket Booking to get hands-on practice on Pega techniques to build an application and to communicate with the other existing applications from the application developed on Pega.

Building a Movie Ticket Booking Application on Pega Platform-

- Team of 4 interns were selected to work on the Movie Ticket Booking Application using Pega Platform.
- I developed data models adhering to Relational Database Management System standards to support the application's functionality. Additionally, I configured portals tailored for various roles within the application.
- I built the application with Organization and Unit names on the Theme Cosmos Architecture of Pega followed by creating the Data Records based on the Data Model.
- Based on the requirements of the workflow from the User Experience Perspective, I built the screen rule using the Section Rule of Pega and also created the Data Pages as well as Report Definition to handle the communication of the Sections with the Data Records.

A main application is given to each and every intern as their Project on which everyone will work individually to complete the frontend, backend and integration of the application on the Pega Platform. I am currently working on the project "Hotel Booking Application" on Pega Platform build Infinity'23.

CHAPTER 5

CHALLENGES AND SOLUTIONS

Throughout my internship at PegaSystems, I encountered challenges that are common experiences for developers, especially when working on a complex tasks.

1. Balancing Frontend and Backend Development

The task I was worked on likely to involved collaboration between frontend developers (focusing on the user interface) and backend developers (handling data and server-side logic). Balancing these two aspects effectively can be a hurdle, especially for an intern:

Solution

1. Regular Communication: To overcome these challenges, I likely adopted a strategy of regular communication. This might have involved daily stand-up meetings, project management tools, or dedicated communication channels to keep both teams informed and on the same page.

2. Code Reviews: Implementing code reviews can be a valuable practice for fostering collaboration and knowledge sharing. By reviewing each other's code, both frontend and backend developers gain a deeper understanding of the overall system and can identify potential issues early on.

3. Knowledge Sharing Sessions: Organizing knowledge-sharing sessions where frontend and backend developers could present their expertise to each other could have been another solution. This promotes cross-team learning and fosters a collaborative environment.

2. Learning New Technologies:

Learning new techniques can be both exciting and challenging, presenting opportunities for growth and innovation. However, it's not uncommon to encounter obstacles along the way. One common challenge is overcoming the initial learning curve associated with unfamiliar concepts or technologies.

Solution

To address this, it's helpful to break down complex topics into smaller, more digestible chunks and to seek out resources such as tutorials, online courses, or mentorship opportunities to gain a solid understanding.

CHAPTER 6

ACHIEVEMENTS AND CONTRIBUTION

1. Gained Knowledge about the Pega technology - Learnt about the Pega terminologies like Dev Studio, App studio, Prediction Studio as well as Admin Studio. - I also learnt about the basics of Pega Platform like rules, section rules, rulesets, data pages, report definition, access groups, personas, etc. - Successfully completed the training period of CSA (Certified System Architect) as well as CSSA (Certified Senior System Architect) in my initial days of internship.
2. Got Certified as a System Architect - Based on the training provided by the company, I successfully cleared the System Architect Certification Exam. - I got certified on the Pega Platform Infinity'23 with a score of 74%. - Started the preparation to clear the Senior System Architect Certification Exam on Pega Infinity'23.
3. Built an application to be well versed with Pega Development - I successfully created the Movie Ticket Booking Application on the Pega Platform Infinity'23 using the App Studio and Dev Studio. - I also learnt about the practical implementation of Pega terminologies which I went through in the training period, and got to know more about what Pega is capable of. I built an Movie Ticket Booking Application that seamlessly integrates email functionality to enhance security measures against unauthorized access to organizational keys
4. Got Certified as a Senior System Architect - Based on the training provided by the company, I successfully cleared the Senior System Architect Certification Exam. - I got certified on the Pega Platform Infinity'23 with a score of 86%.

CHAPTER 7

REFLECTION AND ANALYSIS

REFLECTION

1. Gained Knowledge about the Pega technology: Learnt about the Pega terminologies like Dev Studio, App studio, Prediction Studio as well as Admin Studio. - I also learnt about the basics of Pega Platform like rules, section rules, rulesets, data pages, report definition, access groups, personas, etc. - Successfully completed the training period of CSA (Certified System Architect) as well as CSSA (Certified Senior System Architect) in my initial days of internship

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4. Got Certified as a Senior System Architect: Based on the training provided by the company, I successfully cleared the System Architect Certification Exam. - I got certified on the Pega Platform Infinity'23 with a score of 86%.

ANALYSIS

Impact on Business: The projects described have tangible implications for Pega Systems operations. The Monitoring System provides valuable insights into customer behavior, which can inform decision- making and improve business strategies.

Integration of Technologies: The individual effectively integrates different technologies to develop holistic solutions. For instance, using PostGreSQL for database monitoring demonstrates a synergistic approach to leveraging cloud computing and database management technologies for enhanced performance and reliability.

CHAPTER 8

RECOMMENDATIONS

Continued Learning and Skill Enhancement: Encourage the individual to continue expanding their skill set and staying updated on emerging technologies and industry trends. This could involve pursuing certifications or advanced training programs in areas such as cloud computing, database management, and software development.

Cross-Functional Collaboration: Foster opportunities for the individual to collaborate with colleagues from different departments or teams within PegaSystems. Cross-functional projects can provide exposure to diverse perspectives and approaches, fostering innovation and enhancing problem-solving capabilities.

Leadership Development: Provide opportunities for the individual to take on leadership roles or mentorship responsibilities, allowing them to further develop their leadership skills and inspire others within the organization. This could involve leading project teams, organizing knowledge-sharing sessions, or participating in mentorship programs.

Feedback and Recognition: Offer regular feedback and recognition for the individual's contributions, acknowledging their efforts and achievements. This can boost morale, motivation, and job satisfaction, fostering a positive work environment and encouraging continued growth and performance excellence.

Focus on Impact and Strategic Alignment: Encourage the individual to align their projects and initiatives with the broader strategic goals and priorities of Pega Systems. Emphasize the importance of delivering measurable impact and value to the organization, whether through improved efficiency, cost savings, or enhanced customer satisfaction.

CHAPTER 9

CONCLUSION

As I reflect on my journey at PegaSystems and the significant contributions I've made to the organization, I'm struck by the depth of impact that each project has had. The development of the comprehensive monitoring system using SQL not only demonstrates my technical proficiency but also underscores my ability to derive valuable insights from data to inform strategic decision-making. This project has empowered PegaSystems to better understand customer behavior and market trends, ultimately driving improvements in our products and services.

Moreover, the implementation of an automated system using AWS Lambda showcases my commitment to innovation and efficiency. By leveraging cutting-edge technology, I've streamlined data management processes and enhanced the reliability of our systems, positioning PegaSystems for continued growth and scalability. This project represents a pivotal step forward in our journey towards embracing cloud computing and harnessing its potential to drive business success.

Additionally, the creation of APIs for efficiently fetching bureau data and automating autopay bounce charges underscores my dedication to solving real-world challenges faced by PegaSystems. These solutions have not only optimized our operations but have also improved the overall user experience, fostering greater trust and satisfaction among our customers.

Looking ahead, I'm excited about the opportunities for further growth and development. I'm committed to staying abreast of emerging technologies, fostering cross-functional collaboration, and aligning my efforts with PegaSystems's strategic objectives. By continuing to push the boundaries of innovation and excellence, I'm confident that I can contribute to the continued success and prosperity of PegaSystems in the dynamic and competitive landscape of financial technology.

CHAPTER 10

REFERENCES

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- [2] Pega Community - <https://community.pegasystems.com/>
- [3] Pega Developer Network - <https://community.pegasystems.com/knowledgebase/developer-network>
- [4] Pega YouTube Channel - <https://www.youtube.com/user/PegasystemsInc>
- [5] Pega Documentation - <https://community.pegasystems.com/knowledgebase/products>
- [6] Pega Blogs and Whitepapers - <https://www.pegasystems.com/resources>